

MEMO

WTW158 Calculus

Class Test 3 Mon

1 - 5 March 2025

Time: 20 min

8

Surname:

Initials:

Student number:

Group:

M-number:

Use a pen to complete the test. Simplify all your answers. Use interval notation, where applicable. When a block is provided, write only your final answer in the block. Only this will be marked.

QUESTION 1 [2 marks]

Given $f(x) = \frac{1}{x} - 1$ and $g(x) = \frac{1}{x}$. Determine the following, if defined.

(a) $(g \circ f)(x) =$

$$\frac{x}{1-x} \bigg/ \frac{1}{\frac{1}{x}-1}$$

$$\begin{aligned} (g \circ f)(x) &= g\left(\frac{1}{x}\right) = g\left(\frac{1}{x}-1\right) \\ &= \frac{1}{\frac{1}{x}-1} \\ &= \frac{1}{\frac{1-x}{x}} \\ &= \frac{x}{1-x} \end{aligned}$$

(b) The domain of $g \circ f$ is

$$x \in (-\infty, 0) \cup (0, 1) \cup (1, \infty)$$

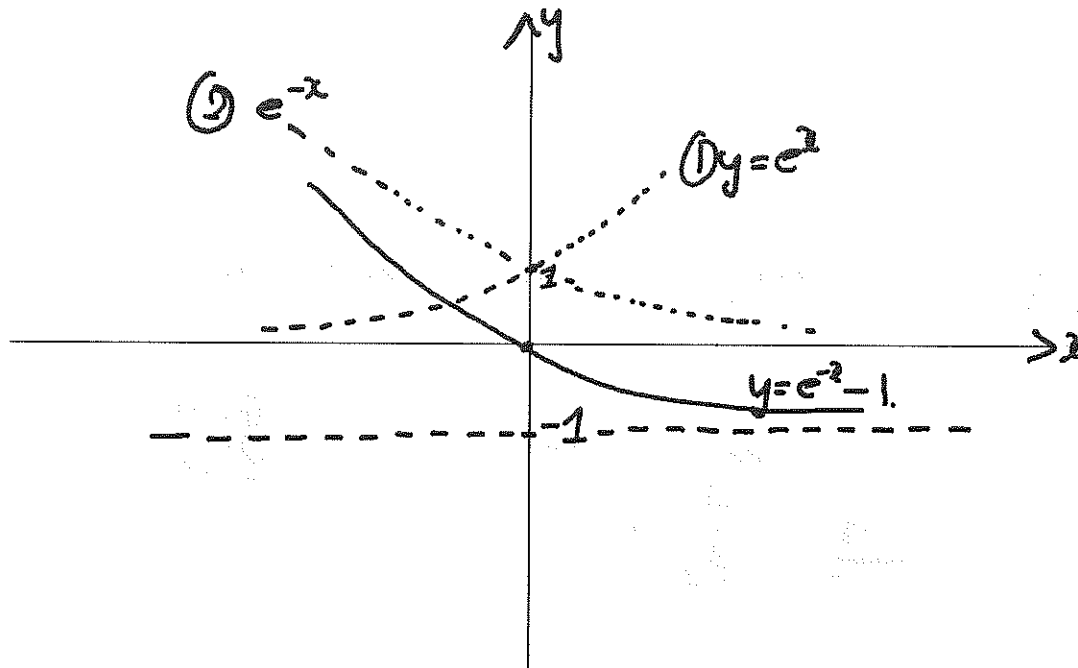
QUESTION 2 [2 marks]

Solve for x if $27^{x-x^2} = \left(\frac{1}{9}\right)^x$

$$\begin{aligned} 27^{x-x^2} &= \left(\frac{1}{9}\right)^x \Rightarrow (3^3)^{x-x^2} = (3^{-2})^x \\ &\Rightarrow 3^{3x-3x^2} = 3^{-2x} \\ &\Rightarrow 3x-3x^2 = -2x \\ &\Rightarrow 5x-3x^2 = 0 \\ &\Rightarrow x(5-3x) = 0 \\ &\Rightarrow x=0 \text{ or } x=\frac{5}{3} \\ &x \in \left\{0, \frac{5}{3}\right\} \end{aligned}$$

QUESTION 3 [2 marks]

By using dotted lines, clearly show how you use transformations of the graph of $y = e^x$, to draw the graph of $y = e^{-x} - 1$. Indicate all necessary detail (asymptotes, intercepts, etc.) on your graph and write down the domain and range of the function $y = e^{-x} - 1$.



QUESTION 4 [2 marks]

Consider

$$f(x) = 1 + \sqrt{x}.$$

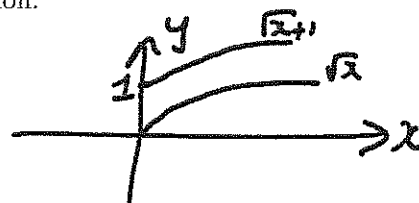
- (a) Find the inverse of the function f .

$$\begin{aligned} y &= 1 + \sqrt{x} \\ \text{let } x &= 1 + \sqrt{y} \\ \Rightarrow \sqrt{y} &= x - 1 \\ \Rightarrow y &= (x - 1)^2 \\ \Rightarrow \underline{f^{-1}(x)} &= (x - 1)^2 \end{aligned}$$

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- (b) Determine the domain and the range of the inverse function.

$$\begin{aligned} D_f &= [0, \infty) = R_{f^{-1}} \\ R_f &= [1, \infty) = D_{f^{-1}} \end{aligned}$$



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Class Test 3 (Tu)

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Use a pen to complete the test. Simplify all your answers. Use interval notation, where applicable. When a block is provided, write only your final answer in the block. Only this will be marked.

QUESTION 1 [4 marks]

Given $f(x) = \sqrt{x}$ and $g(x) = x^2 - 1$. Determine the following, if defined.

(a) $(g \circ f)(x) =$

$x - 1$

(b) The domain of $g \circ f$ is

$[0, \infty)$

(c) The domain of $f \circ g$ is

$(-\infty, 0] \cup [0, \infty)$

(d) $(f \circ g)(-3) =$

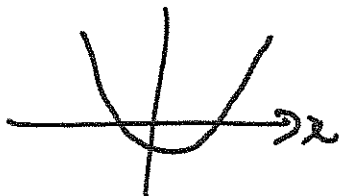
$\sqrt{8} = 2\sqrt{2}$

ROUGH WORK

$$(g \circ f)(x) = g(f(x)) = g(\sqrt{x}) = 1 - 1$$

$$(f \circ g)(x) = f(g(x)) = f(x^2 - 1) = \sqrt{x^2 - 1}$$

$$D_{f \circ g}: x^2 - 1 \geq 0$$



$$(f \circ g)(-3) = f(g(-3)) = f(8) = \sqrt{8} = 2\sqrt{2}$$

QUESTION 3 [4 marks]

Consider the function

$$f(x) = \sqrt{x} + 2.$$

- (a) Find the inverse f^{-1} of f .

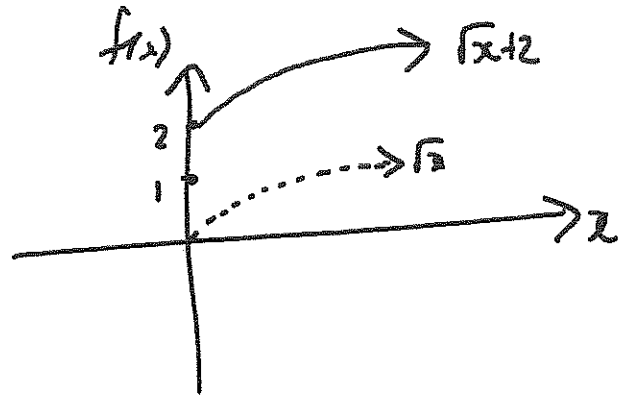
$$y = \sqrt{x} + 2$$

$$\text{let } x = \sqrt{y} + 2$$

$$\Rightarrow \sqrt{y} = x - 2$$

$$\Rightarrow y = (x - 2)^2$$

$$\Rightarrow \underline{f^{-1}(x) = (x - 2)^2}$$

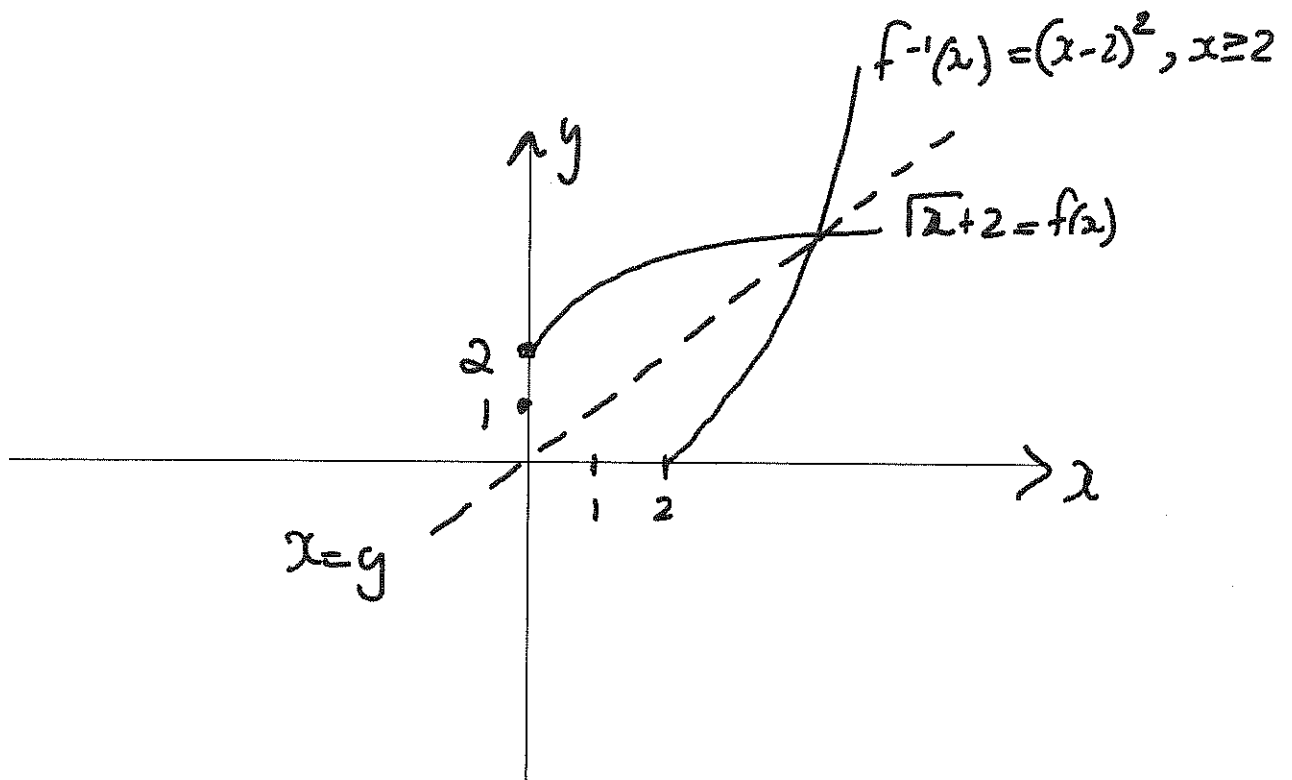


- (b) Determine the domain and the range of f^{-1} .

$$D_f = [0, \infty) = R_{f^{-1}}$$

$$R_f = [2, \infty) = D_{f^{-1}}$$

- (c) Sketch the graphs of f and f^{-1} on the given set of axes and clearly indicate all intercepts and relevant information on the graph.



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Class Test 3 (Th)

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Use a pen to complete the test. Simplify all your answers. Use interval notation, where applicable. When a block is provided, write only your final answer in the block. Only this will be marked.

QUESTION 1 [4 marks]

Given $f(x) = \frac{1}{x}$ and $g(x) = \sqrt{x-3}$. Determine the following, if defined.

(a) $(g \circ g)(x) =$

$$\sqrt{\sqrt{x-3}-3}$$

(b) The domain of $g \circ g$ is

$$[12, \infty)$$

(c) The domain of $f \circ g$ is

$$(3, \infty)$$

(d) $(f \circ g)(12) =$

$$\frac{1}{3}$$

$$\begin{aligned} (f \circ g)(x) &= f(g(x)) \\ &= f(\sqrt{x-3}) \\ &= \frac{1}{\sqrt{x-3}} \end{aligned}$$

ROUGH WORK

$$(g \circ g)(x) = g(g(x)) = g(\sqrt{x-3}) = \sqrt{\sqrt{x-3}-3}$$

Domain of $g \circ g$:

$$\left. \begin{aligned} x-3 &\geq 0, & x &\geq 3 \\ \text{and } \sqrt{x-3}-3 &\geq 0 \\ \sqrt{x-3} &\geq 3 \\ \Rightarrow x-3 &\geq 9 \\ x &\geq 12 \end{aligned} \right\}$$

$$\begin{aligned} (f \circ g)(12) &= f(g(12)) \\ &= f(\sqrt{12-3}) = f(3) = \frac{1}{3} \end{aligned}$$

QUESTION 2 [2 marks]

Solve for x if $\frac{(e^3)^{3-x}}{e^x} = \left(\frac{1}{e}\right)^x$.

$$\frac{(e^3)^{3-x}}{e^x} = \left(\frac{1}{e}\right)^x$$

$$\Rightarrow \frac{(e^3)^{3-x}}{e^x} = \frac{1}{e^x}$$

$$\Rightarrow (e^3)^{3-x} = 1 \Rightarrow e^{9-3x} = e^0 \Rightarrow 9-3x=0 \Rightarrow \underline{x=3}$$

QUESTION 3 [2 marks]

Consider

$$f(x) = \sqrt{x} - 2.$$

Find the inverse f^{-1} of the function f and determine the domain and the range of f^{-1} .

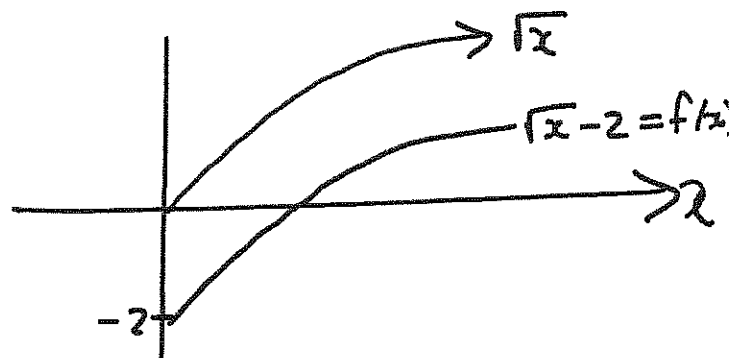
$$y = \sqrt{x} - 2$$

$$\text{let } x = \sqrt{y} - 2$$

$$\Rightarrow \sqrt{y} = x + 2$$

$$y = (x+2)^2$$

$$\Rightarrow \underline{f^{-1}(x) = (x+2)^2}$$



$$D_f = [0, \infty) = R_{f^{-1}}$$

$$R_f = [-2, \infty) = D_{f^{-1}}$$

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Class Test 3 (Fr)

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QUESTION 1 [4 marks]

Given $f(x) = \sqrt{x}$ and $g(x) = x^2 - 1$. Determine the following, if defined.

(a) $(g \circ f)(x) =$

$x - 1$

(b) The domain of $g \circ f$ is

$[0, \infty)$

(c) The domain of $f \circ g$ is

$(-\infty, -1] \cup [1, \infty)$

(d) $(f \circ g)(-3) =$

$\sqrt{8} = 2\sqrt{2}$

$$f(g(-3))$$

$$= f(9-1) = f(8)$$

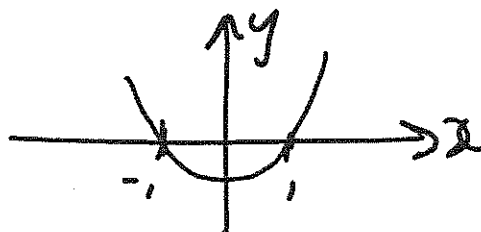
$$(g \circ f)(x) = g(f(x)) = g(\sqrt{x}) = x - 1$$

$$D_{g \circ f} = [0, \infty)$$

$$(f \circ g)(x) = f(g(x)) = f(x^2 - 1)$$

$$= \sqrt{x^2 - 1}$$

$$x^2 - 1 \geq 0$$



QUESTION 2 [2 marks]

Solve for x if

$$8^x = \left(\frac{1}{32}\right)^{x-2}$$

$$(2^3)^x = (2^{-5})^{x-2}$$

$$2^{3x} = 2^{-5x+10}$$

$$3x = -5x + 10$$

$$\Rightarrow 8x = 10$$

$$\Rightarrow x = \frac{10}{8} = \frac{5}{4}$$

QUESTION 3 [2 marks]

Consider the function

$$f(x) = \frac{1}{x} + 2$$

(a) Find the inverse of the function f .

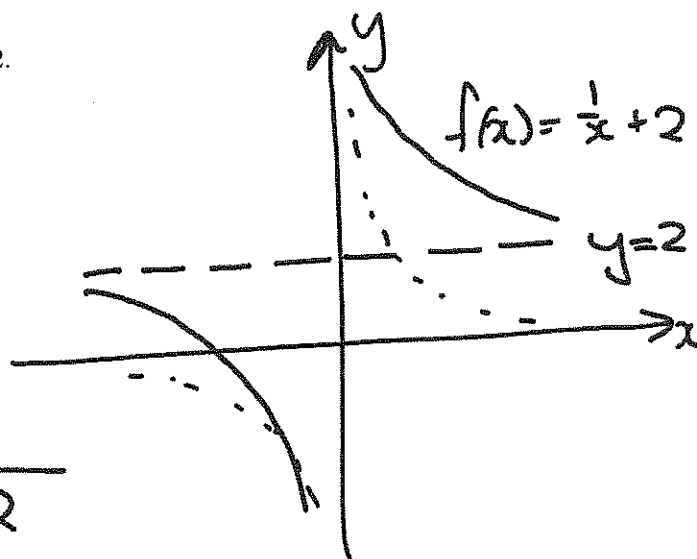
$$y = \frac{1}{x} + 2$$

$$\text{let } x = \frac{1}{y} + 2$$

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$$\Rightarrow \frac{1}{y} = x - 2$$

$$\Rightarrow y = \underline{f^{-1}(x)} = \frac{1}{x-2}$$



(b) Determine the domain and the range of the inverse function.

$$D_f = (-\infty, 0) \cup (0, \infty) = R_{f^{-1}}$$

$$R_f = (-\infty, 2) \cup (2, \infty) = D_{f^{-1}}$$